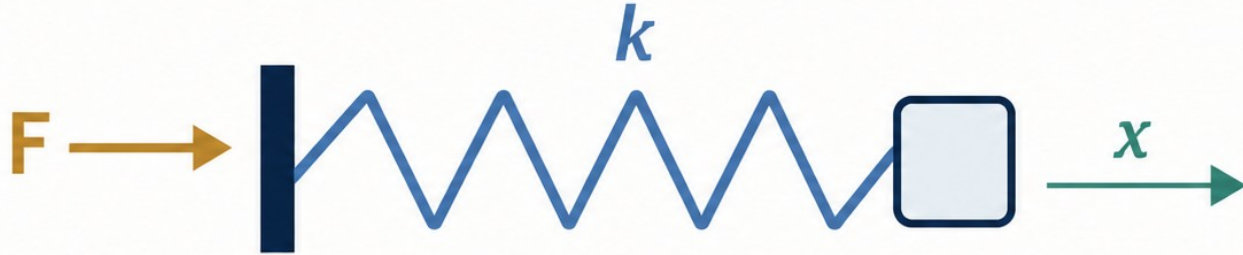


Change one parameter, see the physics

Week 04 | Parameter sweeps and graph interpretation



$$x = F / k$$

Predict before MATLAB

Model: ideal elastic spring (Hooke's law)

$$F = kx, \quad x = \frac{F}{k}$$

Same applied force F

compare extension while k changes



$k = 50 \text{ N/m}$
largest extension



$k = 100 \text{ N/m}$
baseline



$k = 200 \text{ N/m}$
smallest extension

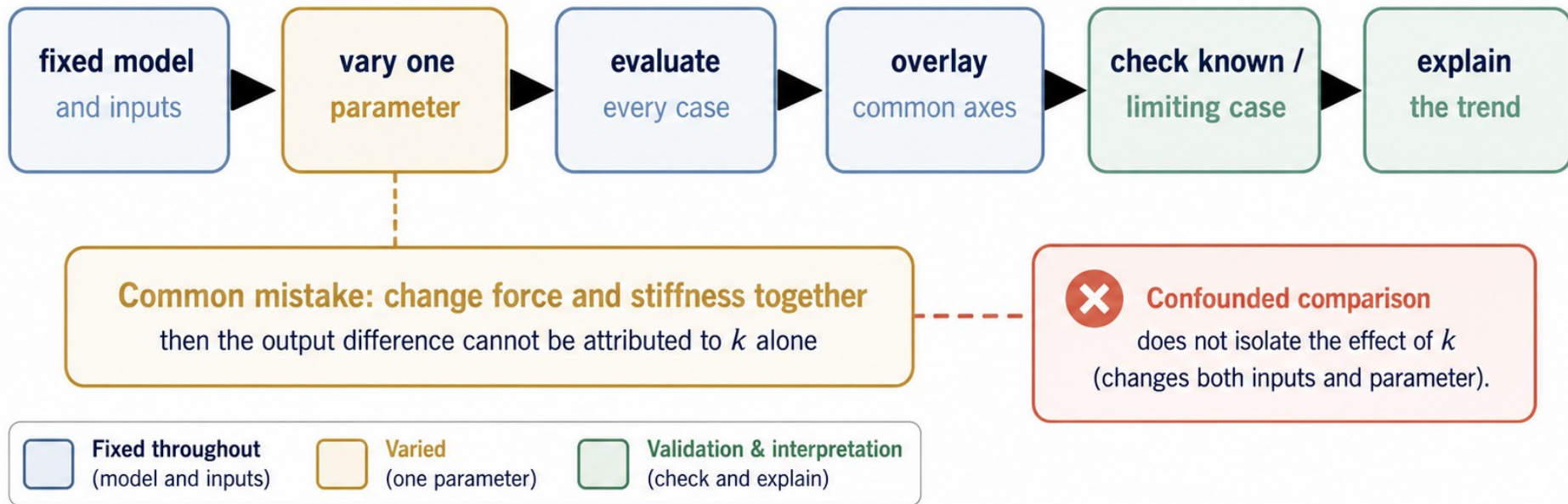


Your predictions

- 1 Hooke's law gives $x = F/k$.
- 2 For the same positive force, **smaller k means larger extension.**
- 3 At $F = 10 \text{ N}$, predict **0.20 m**, **0.10 m**, and **0.05 m** for $k = 50, 100$, and 200 N/m .
- 4 At $F = 0$, every ideal-spring case should give $x = 0$.

A sweep is a controlled experiment

- ✓ Vary one named parameter: spring stiffness k .
- ✓ Hold the force array, model, units, and all other assumptions fixed.
- ✓ A comparison that changes force and stiffness together is confounded.
- ✓ The fixed-input rule lets the output difference be attributed to k .



Start with the model, variables, and units

- ✓ Use an ideal linear spring in its elastic range.
- ✓ $F = kx$, so $x = F/k$.
- ✓ F_N is the common applied-force array in N.
- ✓ k_Npm is the swept stiffness in N/m; extension_m is the output in m.

Physical model: Ideal linear spring

$$F = kx, \quad x = \frac{F}{k}$$

F = applied force (N)

k = spring stiffness (N/m)

x = spring extension (m)

Variables and roles

Variable	Meaning / Role	Unit
F_N	Applied force (common input for all cases)	N
k_Npm	Spring stiffness (swept parameter)	N/m
$x = F/k$	Spring extension (output for each stiffness case)	m

Baseline_k_Npm = 100 N/m (reference case)

Dimensional check

$$\frac{\text{N}}{\text{N/m}} = \text{m}$$

Force (N) divided by
stiffness (N/m) gives
extension in metres (m).



Units are consistent.

Plan the computation in plain language

1		State the model	Use Hooke's law for an ideal spring: $F = kx$ and $x = F/k$.
2		Choose swept parameter	Sweep spring stiffness k (N/m). Define cases: $k = [50 \quad 100 \quad 200]$ N/m.
3		Fix all other inputs	Force array $F_N = 0:0.5:10$ N, equation $x = F/k$, units (N, m), assumptions, and any other inputs remain fixed.
4		Evaluate the same model for every case	For each k case, compute extension $x = F/k$ at every force value.
5		Store results in a 3-by-21 arrangement	One row per stiffness case; one column per force value. Rows correspond to $k = 50, 100, 200$ N/m.
6		Overlay on common axes with legend	Plot spring extension x (m) versus applied force F (N) for all cases on the same axes with labels and legend.
7		Validate a known or limiting case	Check $F = 0$ gives $x = 0$ for every case. (Zero force $\rightarrow$ zero extension.)
8		Explain the trend physically	Higher stiffness $k \rightarrow$ smaller extension x for the same force F . (x is inversely proportional to k at fixed F .)

One controlled change rule

This is a controlled computational experiment.

Change (swept)

Spring stiffness
 k (N/m)



Fixed

- ✓ Force array $F_N = 0:0.5:10$ N
- ✓ Model $x = F/k$
- ✓ Units (N, m)
- ✓ Assumptions
- ✓ Other inputs

Vary one parameter. Hold everything else constant.



Known (limiting) check: $F = 0$ N $\rightarrow x = 0$ m for all k .
Any curve not passing through the origin is physically incorrect.



Common defect to avoid (runnable but physically wrong):

Using $x = F \times k$ instead of $x = F/k$ gives larger x for larger k .
Trend contradicts Hooke's law and physics.

Store the cases explicitly

```
F_N = 0:0.5:10;  
% applied force array (N)  
  
k_Npm = [50 100 200];  
% spring stiffness values (N/m)  
  
baseline_k_Npm = 100;  
% baseline spring stiffness (N/m)
```

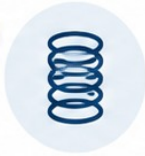
1



Build one common force array.

F_N is the fixed input used for every stiffness case.

2



Store the stiffness cases explicitly.

k_{Npm} lists the swept values 50, 100, and 200 N/m.

3



Keep the baseline named so it can be traced later.

$baseline_k_Npm = 100$ N/m.

4



Units in variable names help keep the physical meaning visible.

N for Newtons, m for metres.

Evaluate the same model for every case

- ✓ Preallocate one row for each stiffness value.
- ✓ Each loop pass changes only $k_{\text{Npm}}(\text{case_id})$.
- ✓ Use element-wise division for $x = F/k$.
- ✓ Row case_id maps to one stiffness case.

MATLAB sweep loop

```
extension_m = zeros(length(k_Npm),length(F_N));  
for case_id = 1:length(k_Npm)  
    extension_m(case_id,:) = F_N./k_Npm(case_id);  
end
```

Result matrix: extension_m (size 3-by-21)

Each row = one stiffness case; each column = one applied force value

Applied force, F_N (N)						$k_{\text{Npm}}(\text{case_id})$ (N/m)
0	0.5	1	1.5	...	10	
0	0.01	0.02	0.03	...	0.20	50 N/m
0	0.005	0.01	0.015	...	0.10	100 N/m
0	0.0025	0.005	0.0075	...	0.05	200 N/m

21 applied-force values: $F_N = 0:0.5:10$ N



Known or limiting case check:

At $F = 0$, every row gives $x = 0$.

This confirms the model and computation are consistent.



Interpretation: Larger stiffness (k) produces smaller extension (x) for the same applied force (F), matching Hooke's law $x = F/k$.

Read one row before plotting

✓ At $F = 10$ N, row 1 gives 0.20 m.

✓ Row 3 gives 0.05 m.

✓ The baseline row gives 0.10 m.

✓ The same-input ordering is $x_{50} > x_{100} > x_{200}$.

	k (N/m)	x at $F = 10$ N (m)	case
1 →	50	0.20	lower stiffness
2 →	100	0.10	baseline
3 →	200	0.05	higher stiffness

Same-input comparison at $F = 10$ N

All three values are for the same applied force. Read down the column to compare the effect of spring stiffness.

$x_{50} = 0.20$ m
(row 1)

>

$x_{100} = 0.10$ m
(baseline)

>

$x_{200} = 0.05$ m
(row 3)

stiffness increases to the right → extension decreases

Overlay every case on common axes

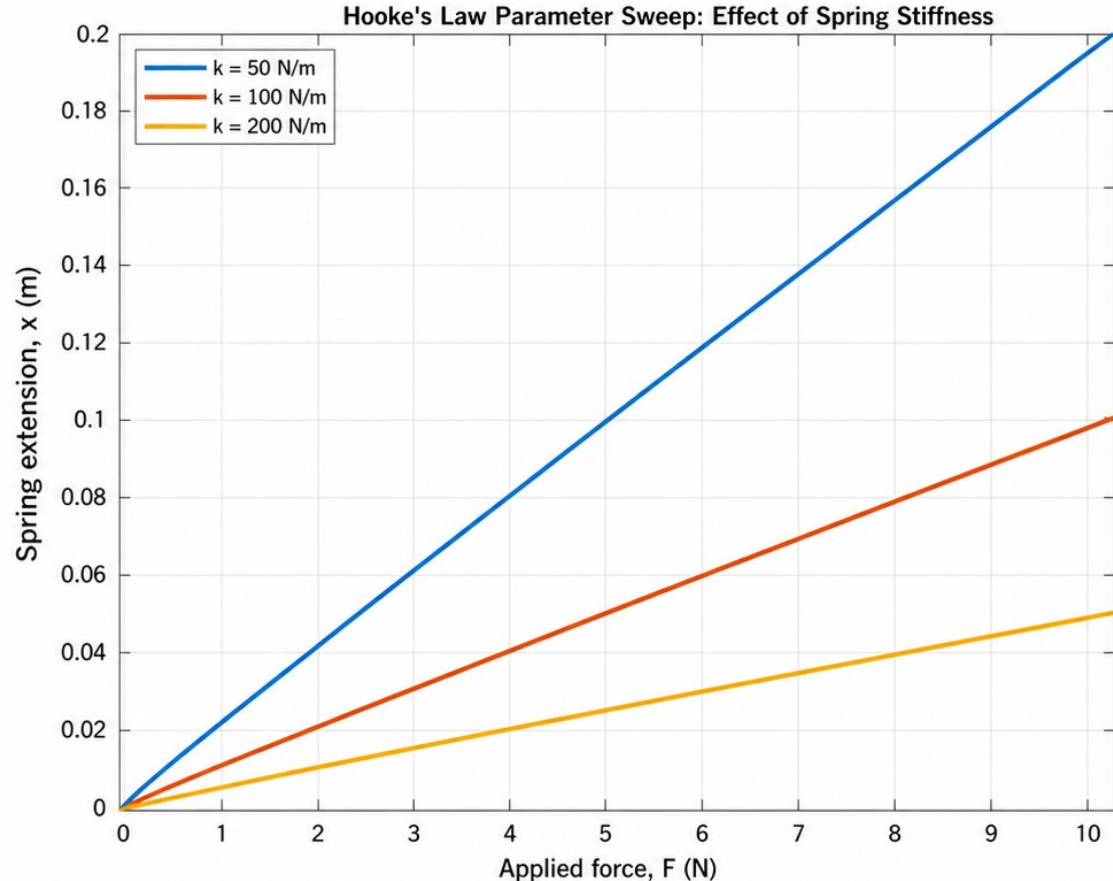
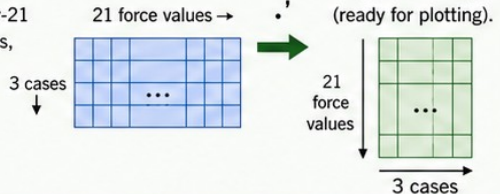
- ✓ Use one plot call for all stiffness cases; label the physical axes and identify each stiffness in the legend.
- ✓ The transpose changes the stored 3-by-21 arrangement into a plotting arrangement without changing the values.

MATLAB plotting command

```
plot(F_N,extension_m.','LineWidth',2)
xlabel('Applied force, F (N)')
ylabel('Spring extension, x (m)')
legend('k = 50 N/m', 'k = 100 N/m', 'k = 200 N/m', ...
       'Location','northwest')
```

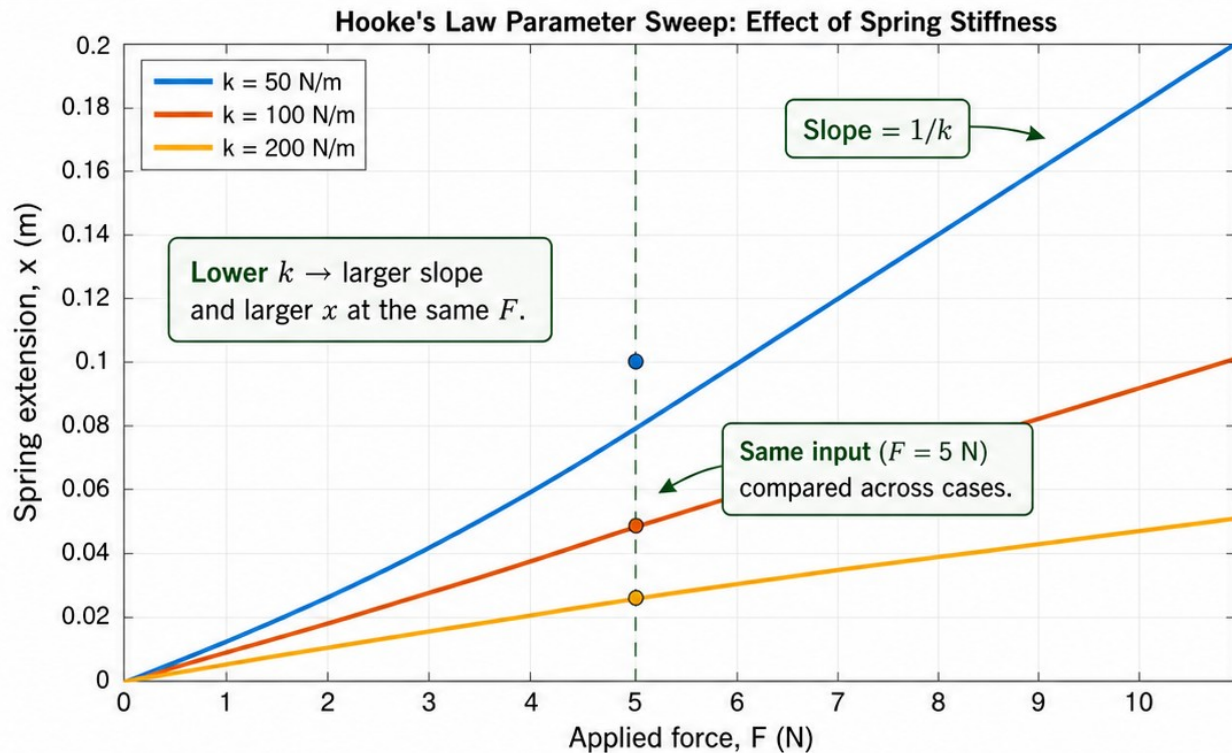
Transpose operation (.)'

Stored result array:
extension_m is 3-by-21
(three stiffness cases,
21 force values).



The graph is a physics statement

- ✓ Each curve is straight because x is proportional to F .
- ✓ The slope is $1/k$.
- ✓ Smaller k gives a steeper line and larger extension at the same force.
- ✓ The graph supports the prediction only because the comparison uses common force inputs.



Physical interpretation: A softer spring (smaller k) stretches more under the same force. A stiffer spring (larger k) resists more and extends less.

$$x = \frac{F}{k}$$

Validate a known limiting case

- ✓ At $F = 0$, Hooke's law requires $x = 0$ for every positive stiffness.
- ✓ Read the first result column directly.
- ✓ Assert that the largest absolute value is below $1\text{e-}12$.
- ✓ A known limit turns a physical expectation into a reproducible check.

MATLAB validation code (strict)

```
zero_force_extension_m = extension_m(:,1)
assert(max(abs(zero_force_extension_m)) < 1e-12, ...
       'Zero-force limiting-case validation failed.')
```

What this tests: The evaluated model output across all cases.

Not a syntax check: It confirms the physics at a known limit.

Zero-force result (first column of extension_m)

At $F = 0$ N,
the spring extension
for every stiffness is
exactly zero.

$\text{extension_m}(:,1) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}$



$\max(\text{abs}(\text{extension_m}(:,1))) = 0 < 1\text{e-}12$

Zero-force limiting-case validation passed.

? **Think:** Why must all three values be zero? What would a nonzero value indicate?

A runnable calculation can still be wrong



DIAGNOSIS: MATLAB runs, but the physics is wrong

Wrong vs correct expressions

```
wrong_baseline_extension = F_N.*baseline_k_Npm;  
correct_baseline_extension = F_N./baseline_k_Npm;  
wrong_final_value = wrong_baseline_extension(end)  
correct_final_value_m = correct_baseline_extension(end)
```

Model being used (unchanged)

$$F = kx, \quad x = \frac{F}{k}$$

Inputs for all cases (fixed)

- Applied force: $F, N = 0:0.5:10$ N
- Stiffness array: $k_Npm = [50 \ 100 \ 200]$ N/m
- Baseline stiffness: $baseline_k_Npm = 100$ N/m



PHYSICAL CHECK: The wrong expression fails

Faulty expression: $x = F.*k$



Unit mismatch

F is in N and k is in N/m.

$N \times (N/m) = N^2/m$,
not metres (m).



Reversed trend

At the same force, larger k gives
larger x in the wrong model.

Correct model $x = F/k$

$k \uparrow \Rightarrow x \downarrow$
(stiffer $\rightarrow$ smaller extension)

Wrong model $x = F.*k$

$k \uparrow \Rightarrow x \uparrow$
(stiffer $\rightarrow$ larger extension)

Repair the expression, then recheck the physics

- ✓ At $F = 10$ N, compare correct $x = F/k$ with wrong $x = F.*k$.
- ✓ The correct model gives 0.20, 0.10, and 0.05 m; the wrong expression gives 500, 1000, and 2000 N²/m.
- ✓ Repair by tracing the model, units, limiting case, and expected trend.

Comparison at $F = 10$ N

k (N/m)	Correct $x = F/k$ (m)	Wrong $x = F.*k$ (N ² /m)
50	0.20	500
100	0.10	1000
200	0.05	2000

Repair path: from suspicion to correction

1

$$F = kx, \\ x = \frac{F}{k}$$

Trace the model

Hooke's law: $F = kx$, $x = \frac{F}{k}$.

2

$$[k] = \text{N/m}$$

Check units

x must be in m.
The wrong output has units N²/m.

3

$$F = 0 \\ \Rightarrow x = 0$$

Apply the limiting case

At $F = 0$, every case must give $x = 0$.

4

$$x \propto \frac{1}{k}$$

Check the expected trend

Larger k should give smaller x .
The wrong expression gives the opposite.

5



Correct the expression

Use element-wise division for $x = F/k$.

```
wrong_baseline_extension = F_N.*baseline_k_Npm;  
correct_baseline_extension = F_N./baseline_k_Npm;  
wrong_final_value = wrong_baseline_extension(end)  
correct_final_value_m = correct_baseline_extension(end)
```

```
correct_baseline_extension = F_N./baseline_k_Npm;
```



Recheck the physics: at $F = 0$, every case gives $x = 0$; at $F = 10$ N, the extensions are 0.20, 0.10, and 0.05 m.

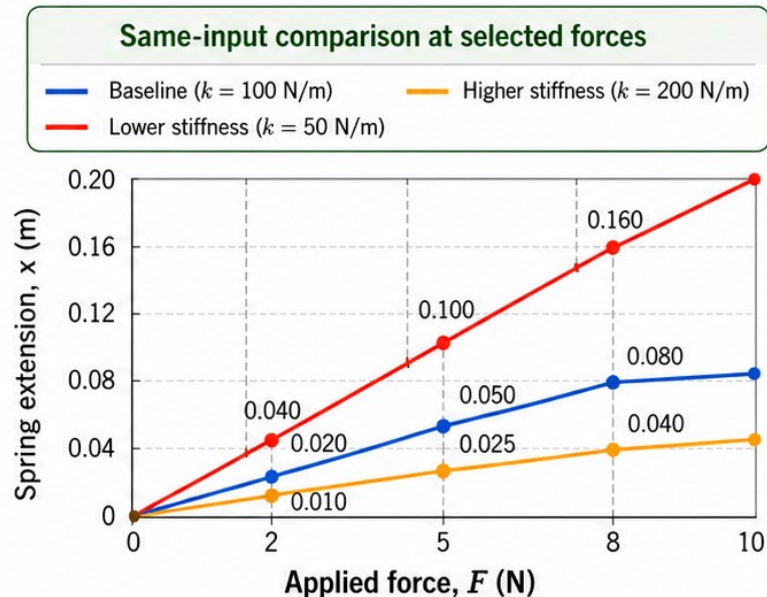
Working exposure: compare selected inputs

- ✓ Compare the same selected forces for every stiffness case.

- ✓ At $F = [2 \ 5 \ 8] \text{ N}$, the values follow the same inverse-stiffness ordering.

- ✓ A table and a graph are two views of the same controlled experiment.

$F \text{ (N)}$	$k = 50 \text{ N/m}$	$k = 100 \text{ N/m}$	$k = 200 \text{ N/m}$
2	0.040	0.020	0.010
5	0.100	0.050	0.025
8	0.160	0.080	0.040



At the same F , extensions follow the extension order:
 $x_{50} > x_{100} > x_{200}$.

Exit ticket — explain a fair sweep to a future you



PREDICT

the trend from the equation.

Hooke's law

$$F = kx \text{ and } x = \frac{F}{k}$$

For a fixed $F > 0$:

$$k \uparrow \longrightarrow x \downarrow$$

Stiffer spring $\rightarrow$ smaller extension.



CONTROL

one changed parameter and the fixed inputs.

Swept parameter:

spring stiffness, k (N/m)

$$k_{Npm} = [50 \ 100 \ 200] \text{ N/m}$$

Fixed model and input array:

Ideal Hooke's law, $x = F/k$

$$F_N = 0:0.5:10 \text{ N}$$

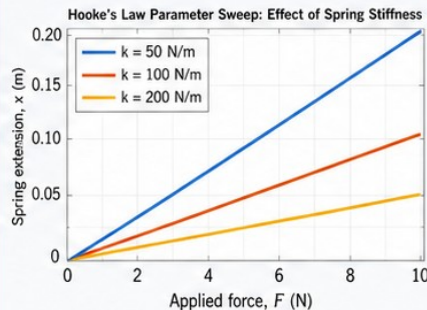
Same model, same F , different k .



PLOT

labelled cases on common axes.

Common axes:



Compare cases at the same F .



CHECK & INTERPRET

a known or limiting case, then the physical result.

Known or limiting case:

At $F = 0$ N, all cases give $x = 0$ m. (Passes.)

Interpretation:

For the same force, a larger k produces a smaller extension. Matches $x = F/k$.

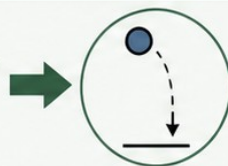


Transfer the workflow

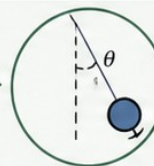
to analogous supplied models.



Ohmic resistor
 $V = IR$



vertical motion
 $s = ut + \frac{1}{2}at^2$



small-angle pendulum

$$T = 2\pi\sqrt{L/g}$$